LESSON PLAN

AIR WEAPON (10)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Study of pressure distribution over an aerofoil at various angles of attack, determination of center of pressure and aerodynamic center.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Study of pressure distribution over an aerofoil at various angles of attack, determination of center of pressure and aerodynamic center.
5. Specific Objectives	10.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer and various concepts of Aerodynamics.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Analysis of pressure distribution data plots to locate CP and AC; Safety: No physical handling of live instruments; Instructor supervision: Monitor data interpretation accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of angle of attack?
(b) How does pressure vary over a symmetrical aerofoil?
(c) Define the boundary layer in the context of flow over a body.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Pressure distribution curves at low angles • Effect of increasing angle of attack • Stall characteristics and flow separation • Calculation of pressure coefficient • Identification of stagnation points	L & Ds	PPT, IB	15 min
(b)	• Definition and movement of Center of Pressure • Shift of CP with varying angle of attack • Definition of Aerodynamic Center • Constant moment property of AC • Comparison of CP and AC locations	L & Ds	PPT, PDF	20 min
(c)	• Graphical determination of CP location • Calculation of AC for standard airfoils • Impact of CP shift on pitching moment • Relevance to aircraft stability • Summary of key aerodynamic concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why the center of pressure moves as the angle of attack changes.
(b) Why is the aerodynamic center preferred over the center of pressure for stability analysis?
(c) Describe the relationship between pressure distribution and the resultant lift vector.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of Center of Pressure and Aerodynamic Center apply to the longitudinal stability requirements discussed in LP 11.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Stability and control: definition, notation used Stability requirements, a/c controls and Maneuvers, static, dynamic stability.	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Sub-Lieutenant, P No. 12345	Senior Instructor, Lieutenant Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min